

Trim line,
see Note 5

1.50"

34.00"

0.75"

See Note 3

GOVERNMENT TITLE BLOCK

See Note 4

Trim line,
see Note 5

See Note 2

NOTES

1. Government will provide contractor with an electronic AutoCAD template to be used by manufacturer for submittal of manufacturer's/contractors drawing in Government format.
2. The manufacturer's title block and drawing revisions as a result of the submittal process, may be located anywhere outside of the trim lines. The location shown on this drawing is the Government's preferred area.
3. The manufacturer shall show the Government solicitation/specification number, the manufacturer's name, drawing and sheet number in this area.
4. Government drawing number(s) will be issued at the end of the submittal process and drawing will be entered into the Government's electronic drawing system by the Government.
5. Trim lines shall be shown on all submitted drawings.

REFERENCE

1. Information Management Handbook (IMH - 2020), Chapter 2, Section 9.A.4 - "Sheet Border" and Figures G2, G3, and G4.



ALWAYS THINK SAFETY

U.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
YAKIMA PROJECT
KENNEWICK DIVISION

CHANDLER PUMPING PLANT

SWITCHYARD - STEEL STRUCTURES

SAMPLE TITLE

DRAWING BORDER

MANUFACTURER OR CONTRACTOR
DRAWN

ACCEPTED

TITEL

SAMPLE SIGNATURE LINES

— BUREAU OF —
RECLAMATION

THINK SAFETY

WAYS THINK SAFE

J.J.S. DEPARTMENT OF THE INTERIOR
BUREAU OF RECLAMATION
STANDARD DRAWINGS

STANDARD AND TYPICAL DESIGNS

GOVERNMENT DRAWING FORMAT

REV NO	2023-04-20	D	UPDATED DRAWING TO CURRENT DRAWING RECORDS SYSTEM. MINOR REVISIONS.
3			

REV NO	2015-03-25	D	UPDATED TITLE BLOCK AND SIGNATURE LINES.
--------	------------	---	--

2	Barthel, Rodney J P.E.	
REV NO	2014 12 17	0
UNDATED TITLE BLOCK AND SIGNATURE LINES		

1	E. Mendlin	2014-12-17	UPDATED TITLE BLOCK AND SIGNATURE LINES.
---	------------	------------	--

S. NITTA

DESIGNED
S. NITTA

3. NITTA
DRAWN

D. CRAW
CHECKED

R. BARTI

TECH. APPR.

R.W. LAF
APPROVED

ADMIN APPROV _____

DENVER, C

DRAWING BORDER,
SAMPLE TITLE BLOCK,
AND SIGNATURE LINES

40-D-7102

SHEET 1

CAD SYSTEM: 22.06
CADD AND PLOT: 22.06
CADD FILENAME: 40-D-60003-RO-L.DWG
DATE AND TIME PLOTTED: MAY-27-2020 13:55
PLOTTER BY: RODRIGUEZ

Released

D

C

B

A

1

2

3

4

5

ABBREVIATIONS

add'l = additional
bf = bottom face
bl = bottom layer
br = bottom row
CJ = Construction joint
CL = Centerline
CLr. = clear
Cr.J = Contraction joint
Ct.J = Control joint
ctr. = center or centers
d_b = nominal diameter of reinforcing bar
Dwg. = Drawing
ec = each corner
ef = each face
EJ = Expansion joint
El. = Elevation
eq. = equal or equally
eq. spc. = equally spaced, equal spaces
er = each row
es = each side
ew = each way
ff = far face
fr = far row
fs = far side
HP = High point

if = inside face
ir = inside row
l_d = development length
l_{dh} = development length of standard hook
LP = Low point
ml = middle layer
mr = middle row
nf = near face
nr = near row
ns = near side
OCJ = Optional construction joint
of = outside face
or = outside row
spc. = spaced or spaces
Std. = Standard
tf = top face
tl = top layer
TOC = Top of concrete
tr = top row
TW = Tailwater
Typ. = Typical
uv = uniformly varying lengths of bars between lengths shown
VCJ = Vertical construction joint
WP = Working point
WS = Water surface, Waterstop

CONCRETE SYMBOLS

The different concrete placements are indicated by the following symbols:

	Concrete - First stage
	Concrete - Second stage
	Concrete - Blockout
	Concrete - Existing or in adjacent structures
	Concrete - Precast
	GROUT

NOTES TO DESIGNERS AND DETAILERS

1. Splice lengths shown in TABLES 3.0-60 thru 5.0-60 on 40-D-60004 are for Class B tension lap splices in accordance with ACI 318 and ACI 350. Assumed conditions for these tables in addition to the requirements shown on this drawing are uncoated reinforcement, normal weight concrete, reinforcement yield strength fy = 60,000 psi, and the transverse reinforcement index (K_{tr}) equal to zero. Splices or development lengths other than those shown in the tables must be detailed on the reinforcement specifications drawings.

2. Some factors which require additional consideration are: Beams or columns with ties, lightweight aggregate concrete, epoxy-coated reinforcement, excess reinforcement, bars in compression, bundled bars, and special seismic provisions.

3. The minimum centerline to centerline bar spacing requirements shown in TABLES 3.0-60 thru 5.0-60 on 40-D-60004 may not be applicable when using the provisions of ACI 350 Provision 10.6 for distribution of flexural reinforcement to control flexural cracking.

4. The minimum centerline to centerline bar spacing requirements shown in TABLES 3.0-60 thru 5.0-60 on 40-D-60004 is based on using 1.5 inch aggregate to calculate the spacings.

5. Recalculation of lap splice lengths and development lengths by the designer may be required for changes in combinations of controlling variables per ACI 318 Provisions 25.4 and 25.5, or ACI 350 Provisions 12.8 and 12.9.

6. Lap splices and development lengths for horizontal bars in walls are designated as top bar splices and development lengths unless otherwise noted or detailed on the reinforcement specifications drawings.

7. Development lengths for deformed bars in tension terminating in a standard hook shall be calculated by the designer in accordance with ACI 318 Provision 25.4.3.1 or ACI 350 Provision 12.8.5. In cases where a restricted member thickness does not allow for the full development length of a standard hook, the designer shall use a reduced tensile capacity.

CONCRETE OUTLINE GENERAL NOTES

CHAMFER :
Chamfer edges of permanently exposed concrete surfaces, except slabs and top edges of curbs, with a 45° bevel, $\frac{3}{4}$ " x $\frac{3}{4}$ ", unless otherwise shown on the specifications drawings.

CONCRETE PLACEMENT :
Before placing concrete, see specifications and manufacturer's drawings, for all embedded material which is required in the placement.

DIMENSIONS :
All dimensions to a joint are to the centerline of the joint unless otherwise shown.
Dimensions to beams, columns, and walls are from reference lines or other control points.
Beam and slab depths shall be measured from the top of the structural slab.
Dimensions given for the depth of recesses are from the surface of the structural concrete.
Thicknesses shown for walls and slabs placed against soil or rock are minimum dimensions.

EMBEDDED MATERIALS :
Before placing concrete, care shall be taken that all embedded items are in position and securely fastened in place. Anchor bolt locations and settings are shown on the structural steel, miscellaneous metal, concrete, and equipment drawings. Conduits and grounding cables are shown on the manufacturer's drawings and the electrical drawings. All projecting items, including waterstops, shall be supported and protected from damage and exposure.

FINISHES :
For location and classes of finishes for formed and unformed concrete surfaces, see the specifications drawings and/or paragraphs.

JOINTS :
All construction joints, identified as CJ, shall be provided where shown on the drawings. All optional construction joints, identified as OCJ, are to reduce the volume of placements or to facilitate construction. If optional joints are provided they shall be only at the locations shown. Relocation, addition, or elimination of construction joints will be subject to approval.

TOOLED EDGES :
Tool exposed edges of slabs and top edges of curbs to a radius of $\frac{1}{4}$ ", unless otherwise shown on the specifications drawings.

REINFORCEMENT GENERAL NOTES (See Note 4)

ACCESSORIES :
Bar supports, spacers, and other accessories are not shown on the design drawings. The recommendations of the ACI Detailing Manual-2020 (MNL-66), or other approved supporting systems shall be used.

CODE AND DETAILING REFERENCES :
ACI Building Code Requirements for Structural Concrete (ACI 318-19).
ACI Detailing Manual-2020 (MNL-66).
ACI Code Requirements for Environmental Engineering Concrete Structures (ACI 350-20).

COVER :
Place the reinforcement so that the clear distance between face of concrete and nearest reinforcement is 2" for #8 bars and smaller, and 3" for #9 bars thru #11 bars, unless otherwise shown on specifications drawings. Provide 3" clear distance from face of concrete for all bars when concrete exposure is cast against and permanently in contact with the ground. For corner bars cover, see ADDITIONAL REINFORCEMENT AROUND OPENINGS AND RECESSES, SECTION A-A on 40-D-60004. Clear distance is to the design dimension line.
Reinforcement parallel to construction joints shall have a minimum of 2" clear cover.

DIMENSIONS :
Dimensions are to the centerline of the bars except for embedment of hooks, which are dimensioned to the outside of the bar.
Clear cover dimensions are marked "clr." and are dimensioned to the outside of the bar.

PLACING :
Reinforcement at small openings (Max. 1'-6") in walls and slabs may be spread apart not more than 1.50 times the bar spacing. Unless otherwise shown on the specifications drawings, reinforcement shall be adjusted laterally to maintain a clear distance of at least 1" between the reinforcement and keys, waterstops, anchor bolts, form ties, conduits, and other embedded materials. In heavily reinforced areas, relocation of the embedded material or cutting of reinforcement must be approved.
When bars are bent due to offsets less than 3" and recesses less than 3" deep, the slope of the inclined portion must not exceed 6 to 1. Reinforcement parallel to anchor bolts or other embedded material shall be placed to maintain a clear distance of at least 1.33 times the maximum size aggregate.

PLAIN DOWELS :
Plain dowels across contraction joints shall be smooth bars uniformly coated with a film of oil before concrete placement.

REINFORCEMENT DOWELS :
Unless otherwise shown, dowels indicated on the drawing, such as #8(d), shall be embedded a length equal to l_d and shall have a projection equal to that required for lap splicing to a bar of the same diameter.

SPACING :
The first and last bars in walls and slabs, stirrups in beams, and ties in columns are to start and end at a maximum of one half of the adjacent bar spacing. The minimum edge spacing shall be 2.5d_b. For #9 thru #11 bars where minimum centerline bar spacing is 6" and the lap splice lengths requiring staggered placement shown in TABLES 3.0-60 thru 5.0-60 on 40-D-60004 are used, the minimum edge spacing shall be 3".

SPICES :
The minimum length of lap for splicing parallel bars in tension shall be as given in TABLES 3.0-60 thru 5.0-60 on 40-D-60004. Staggered lap splices shall be separated to give 12 inches clear between ends of adjacent splices.
Bars spliced by noncontact lap splices shall not be spaced transversely farther apart than one-fifth the required lap splice length, nor 6" on centers.
When reinforcing bars of different size are to be spliced, splice length shall be the larger of l_d of larger bar and tension lap splice length of smaller bar.
Splices are to be made so that the required clear distances to face of concrete will be maintained.

STANDARD HOOKS FOR DEVELOPMENT OF DEFORMED BARS IN TENSION :
90-degree bend plus 12d_b extension at free end of the bar.
180-degree bend plus 4d_b extension, but not less than 2 $\frac{1}{2}$ ", at the free end of the bar.

STANDARD HOOKS FOR STIRRUPS, TIES, AND HOOPS :
#5 bar and smaller, 90-degree bend plus 6d_b extension, but not less than 3", at the free end of the bar.
#6, #7, and #8 bars, 90-degree bend plus 12d_b extension at the free end of the bar.
#8 bars and smaller, 135-degree bend plus 6d_b extension, but not less than 3", at the free end of the bar.
#8 bars and smaller, 180-degree bend plus 4d_b extension, but not less than 2.5", at the free end of the bar.

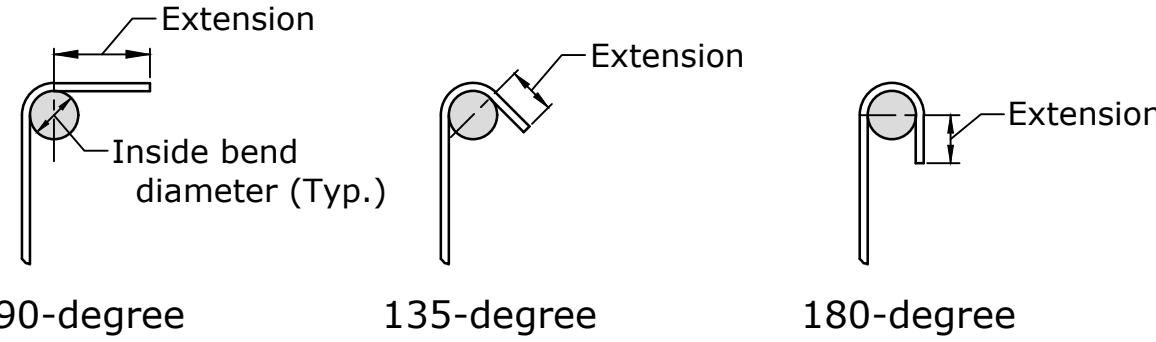


TABLE OF INSIDE BEND DIAMETERS (INCHES)

BAR NO.	3	4	5	6	7	8	9	10	11
Standard	2 $\frac{1}{4}$	3	3 $\frac{3}{4}$	4 $\frac{1}{2}$	5 $\frac{1}{4}$	6	9 $\frac{1}{2}$	10 $\frac{3}{4}$	12
Stirrups, ties, and hoops	1 $\frac{1}{2}$	2	2 $\frac{1}{2}$	4 $\frac{1}{2}$	5 $\frac{1}{4}$	6			

SYMBOLS :
Bars shown thus:
 #8@1-0 or #6@8
Indicate a group of the same size bars equally spaced.
 An open circle at the end of a bar indicates a bend with the bar turned away from the observer.
 A closed circle at the end of a bar indicates a bend with the bar turned towards the observer.
Splices shown thus indicate a lap splice, not a bend in the bar.
 Mechanical splice
 Headed deformed bar

NOTES

1. This drawing shall be used in conjunction with 40-D-60004.

2. For Minimum Requirements For Detailing Reinforcement, see 40-D-60004.

3. Unless otherwise shown on the concrete outline and/or reinforcement specifications drawings, the notes shown are minimum requirements and typical for all drawings that refer to this drawing.

4. Unless otherwise shown on the reinforcement specifications drawings or 40-D-60004, follow the recommendations established by the ACI Detailing Manual - 2020 (MNL-66).

5. # Denotes reinforcement bar number.

6. Numerals in parentheses () following notes, details, and section letters or numbers indicate the number of the drawing upon which the section or detail is indicated or shown. For example (60524) denotes drawing XXX-D-60524. The letters XXX shown here will be numbers denoting the project.

REV NO	2022-09-21	2020-05-20
DESIGNED	Bartel, Rodney J P.E.	Bartel, Rodney, P.E.
DRAWN		
CHECKED		
TECH. APPR.	McGovern, Robert K P.E.	
APPROVED	Bernstein, Alfred I	
FOR REVIEWER -	Plant Structures Group	
DENVER, CO		2016-09-23

GENERAL NOTES FOR
CONCRETE OUTLINE AND
REINFORCEMENT

40-D-60003

SHEET 1 OF 2

THIS DRAWING SUPERCEDES DRAWINGS 40-D-6263 AND 40-D-7012

